

Some studies related to Fermat's Last Theorem

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SETUP AND MINIMUM VALUES

[0] $a^n + b^n = c^n$ where $a < b < c$ and a, b, c , and n are positive integers with $n \geq 2$
since $c > a$ and $c > b$, c can be written as $x+a$ or $y+b$ where x and y are positive integers

[1] thus $a^n + b^n = (x+a)^n = (y+b)^n$

now expand $(x+a)^n$ and $(y+b)^n$:

$(x+a)^n = x^n + a^n + \text{remaining terms (call their sum "f")}$

$(y+b)^n = y^n + b^n + \text{remaining terms (call their sum "g")}$

thus $a^n + b^n = x^n + a^n + f = y^n + b^n + g$

subtract from both sides

$b^n = x^n + f$

$a^n = y^n + g$

f and g are both positive integers, thus $b^n > x^n$ and $a^n > y^n$

take n th root of both sides

$b > x$ and $a > y$

write b as $x+s$ and a as $y+r$ where s and r are positive integers

now back to [1]

$a^n + b^n = (x+a)^n = (y+b)^n$

$a^n + b^n = (x+(y+r))^n = (y+(x+s))^n$

$a^n + b^n = (x+y+r)^n = (x+y+s)^n$

thus $r = s$; rename this integer Z .

[2] then $a^n + b^n = (x+y+Z)^n$

by definition, $a \neq b$ and since $c = x+a = y+b$, $x \neq y$

[3] $y+Z = a$ and $y < a$, thus **a must be at least 2** (if $y = Z = 1$)

$b > a$ so **b must be at least 3**

[4] thus $a + b \geq 5$

now back to [2]:

$a^n + b^n = (x+y+Z)^n$

recall that $b = x+Z$ and $a = y+Z$. thus $x = b-Z$ and $y = a-Z$. Substitute these values:

$a^n + b^n = ((b-Z)+(a-Z)+Z)^n$

[5] $a^n + b^n = (a + b - 2Z + Z)^n = (a + b - Z)^n$

Suppose we imagine that $(a, b, Z) = (2, 3, 1)$ works for some unknown integer value of n . Substituting these values of a, b , and Z into [5] gives

[6] $2^n + 3^n = (2 + 3 - 1)^n = 4^n$

Divide both sides by 4^n

$$\left(\frac{1}{2}\right)^n + \left(\frac{3}{4}\right)^n = 1$$

Both terms on the left are less than 1 and get smaller as n increases, thus n must be small. Substituting $n = 1$ gives $\left(\frac{1}{2}\right) + \left(\frac{3}{4}\right) = 1.25$ which is greater than 1

Substituting $n = 2$ gives $\left(\frac{1}{4}\right) + \left(\frac{9}{16}\right) = 0.8125$ which is less than 1

All larger values of n will give a value for $\left[\frac{1}{4}\right]$ that is less than 0.8125, since both exponent terms are decreasing

Thus n must be between 1 and 2 if $a = 2$, $b = 3$, and $Z = 1$. But this violates our initial condition that n is a positive integer

[7] Thus $a = 2$, $b = 3$, and $Z = 1$ cannot be correct

Is Z wrong?

Recall that $b = x+Z$ and $a = y+Z$, where $c = x+a = y+b$. We know $1 \leq x$ and $1 \leq y$, so $1 \leq b - Z$ and $1 \leq a - Z$. Thus $Z \leq a - 1$ and $Z \leq b - 1$. If our current values $a = 2$ and $b = 3$ have a solution then

$$Z \leq a - 1$$

$$Z \leq 2 - 1$$

$$Z \leq 1$$

So there are no other possible values of Z for $a = 2$ and $b = 3$, and the only one available, $Z = 1$, leads to a value of n that violates initial conditions. Thus either $a = 2$, $b = 3$, or both cannot satisfy $[0]$, which means that either $a \geq 3$, $b \geq 4$, or both must be true. Since the minimum value of either a , b , or both increases by 1 regardless, we know that $a + b \geq 2 + 3 + 1$ or **$a + b \geq 6$** .

Suppose $a + b = 6$. Then we have

$$a^n + b^n = (a + b - Z)^n$$

$$a^n + (6 - a)^n = (6 - Z)^n$$

We know that $Z \leq a - 1$, so $(6 - Z)^n \geq (6 - (a - 1))^n$. Thus

$$a^n + (6 - a)^n \geq (7 - a)^n$$

Divide by $(7 - a)^n$

$$\left(\frac{a}{7 - a}\right)^n + \left(\frac{6 - a}{7 - a}\right)^n \geq 1$$

Suppose we assume $a = 2$ (still our current minimum value for a). Then

$$\left(\frac{2}{5}\right)^n + \left(\frac{4}{5}\right)^n \geq 1$$

Since both bases on the left hand side are fractions < 1 , the largest possible value for the left hand side (with $n \geq 2$) will be when $n = 2$. Testing with this value gives

$$\left(\frac{4}{25}\right) + \left(\frac{16}{25}\right) = \frac{20}{25} < 1$$

So $a = 2$ cannot be a solution. Now try $a = 3$.

$$\left(\frac{a}{7 - a}\right)^n + \left(\frac{6 - a}{7 - a}\right)^n \geq 1$$

$$\left(\frac{3}{4}\right)^n + \left(\frac{3}{4}\right)^n \geq 1$$

$$2 \cdot \left(\frac{3}{4}\right)^n \geq 1$$

$$\left(\frac{3}{4}\right)^n \geq 0.5$$

Again, the largest possible value for the left hand side (with $n \geq 2$) will be when $n = 2$. Testing with this value gives

$(9 / 16) \geq 0.5$, so the function evaluates correctly when $n = 2$. Now try $n = 3$.

$(27 / 64) \geq 0.5$, so the function evaluates incorrectly when $n = 3$. All larger values of n will lead to even smaller values on the left hand side, so only $n = 2$ works when $a = 3$.

However, if $a = 3$ when $a + b = 6$, then $b = 3$. But we know $b > a$. And if $a > 3$, then b must be less than 3 for $a + b$ to equal 6, which also violates the initial condition that $b > a$. Thus there are no workable values of a , and **$a + b = 6$ cannot be a solution to [0]**.

[8] Therefore, **$a + b > 6$ or $a + b \geq 7$, and $a \geq 3$ and $b \geq 4$.**

What about Z ? For the Pythagorean solution $3^2 + 4^2 = 5^2$, $Z = (3+4) - 5 = 2$. Can Z ever be smaller than this? That is, can Z ever equal 1?

We know $a + b \geq 7$. Subtract 1 from both sides

$$a + b - 1 \geq 7 - 1$$

Raise to the power of n

$$(a + b - 1)^n \geq 6^n$$

[9] Imagine that $Z = 1$. Then the left hand side can be written as $(a + b - Z)^n$.

$$(a + b - Z)^n \geq 6^n$$

Substitute from [5]:

$$a^n + b^n \geq 6^n$$

We know $b > a$ and $b^n > a^n$, so we can write

$$2 * b^n \geq a^n + b^n \geq 6^n, \text{ or simply}$$

$$2 * b^n \geq 6^n$$

Take n th root of both sides

$$2^{(1/n)} * b \geq 6$$

$$2^{(1/n)} \geq 6 / b$$

The largest possible value of $6 / b$ is $6 / 4 = 1.5$. So

$$2^{(1/n)} \geq 6 / 4 \geq 6 / b$$

$$2^{(1/n)} \geq 1.5$$

But for $n \geq 2$, the maximum value of $2^{(1/n)}$ is $1.4142... \geq 1.5$

Thus the initial assumption of $Z = 1$ from [9] cannot be correct.

[10] So **Z must be at least 2**

FINDINGS ABOUT THE SYSTEM

$$a^n + b^n = c^n = (x + a)^n = (y + b)^n = (a + b - Z)^n = (x + y + Z)^n = (x + Z)^n + (y + Z)^n$$

$$c = a + b - Z = x + y + Z$$

$$x + Z = c - y$$

$$a - Z = c - b$$

$$a \geq 3$$

$$b \geq 4$$

$$x \geq 2$$

$$y \geq 1$$

$$Z \geq 2$$

$$a = y + Z$$

$$b = x + Z$$

$$x = b - Z$$

$$y = a - Z$$

$$b > x$$

$$a > y$$

$$b > a$$

$$b > y \text{ (follows from previous two)}$$

$$c < b \cdot \sqrt{2} \quad (\text{p. 116})$$

$$y / b < \sqrt{2} - 1 \quad (\text{p. 116})$$

$$x > y, \text{ because } b > a \text{ and thus } x + Z > y + Z$$

$$b - a = x - y \text{ so } b < x - y$$

$$a + b \geq 7$$

$$ab \geq 12$$

$$x + y \leq 2$$

$$x + y \geq 3$$

$$x + y = a + b - 2Z \text{ so } x + y + 2Z = a + b \text{ and } x + y < a + b$$

$$x + y < a + b$$

$$3 \leq a + b - 2Z$$

$$9 - (a + b) \leq Z \leq a + b - 5$$

$$b - a = x - y$$

$$b - x = a - y$$

$$x + a = b + y$$

$$(x + a) = (x + y + a + b) / 2 = (y + b)$$

$$x + a = c$$

$$y + b = c$$

(the above two because it's defined so at the very beginning of the proof; but also, $x + a = b + y$ and $c = (a + b + x + y) / 2$)

$$x + a \geq 5$$

$$b + y \geq 5$$

$$x + a > 2y$$

$$x + a < 2b$$

$$x < 2b - a$$

$$c = (a + b + x + y) / 2$$

$$c + Z = a + b$$

$$c - Z = x + y$$

$$Z = a + b - c = c - x - y$$

$$Z = (a + b - x - y) / 2$$

$$z - 1 \geq (a + b - c) / 2 \geq 1$$

$$z \geq (a + b - c + 2) / 2 \geq 2$$

$$b \leq (a^n + b^n)^{1/n} \leq b \sqrt[n]{2} \quad (\text{p. 116})$$

$$b \leq (a^n + b^n)^{1/n} \leq ((a + b)^n - 2^* a^* b)^{1/n} \quad (\text{p. 116})$$

$$(a^n + b^n)^{1/n} \geq (3^n + b^n)^{1/n} \geq (9 + b^n)^{1/n}$$

$x + y + 2 \leq (a^n + b^n)^{1/n} \leq a + b - 2$ (all values of the center term for $n=2$ seem to be exactly halfway btwn the upper and lower bounds)

$(x - b + 5)^n \leq a^n + b^n \leq (x + 2a + b - 5)^n$ (p. 104 top) (like the the originally found bounds for Z , not super helpful - lower bound often negative, upper bound enormous)

More inequalities or equivalences (p. 108):

$(x + y - 2)^n$	$\leq a^n + b^n \leq$	$(a + b + 2)^n$	
$(x + y - Z)^n$	$\leq a^n + b^n \leq$	$(a + b + Z)^n$	
$(c - Z)^n$	$\leq a^n + b^n \leq$	$(c + Z)^n$	
$((a+b+x+y)/2 - Z)^n$	$\leq a^n + b^n \leq$	$((a+b+x+y)/2 + Z)^n$	
$(x + a - Z)^n$	$\leq a^n + b^n \leq$	$(x + a + Z)^n$	but $x + a = c$
$(y + b - Z)^n$	$\leq a^n + b^n \leq$	$(y + b + Z)^n$	but $y + b = c$
or $(P - Z)^n$	$\leq a^n + b^n \leq$	$(P + Z)^n$	
$(x + a - Z)^n$	$\leq (x + a)^n \leq$	$(x + a + Z)^n$	
$(y + b - Z)^n$	$\leq (y + b)^n \leq$	$(y + b + Z)^n$	
$(x + y)^n$	$\leq a^n + b^n \leq$	$(a + b)^n$	
$(x + y)^n$	$\leq (x + a)^n \leq$	$(a + b)^n$	
$(x + y)^n$	$\leq (y + b)^n \leq$	$(a + b)^n$	
$(x + a)^n$	$= ((x+y+a+b)/2)^n =$	$(y + b)^n$	

or simply $(x + a) = (x+y+a+b)/2 = (y + b)$

Also note that, at least for all Pythagorean triples I've examined, $a + b$ and $x + y$ appear to be equidistant from c . So there must be a number N where $c - N = x + y$ and $c + N = a + b$. **That number is

Z ...duh...because $a + b - Z = x + y + Z = c$. Guess I just never thought to write it as

$c - Z = x + y$ and $c + Z = a + b$.

Some variable values when $n = 2$ (rightmost 3 columns are the sums and products of digits)

a	b	c	x	y	Z	a ²	b ²	c ²	+/*	+/*	+/*
3	4	5	2	1	2	9	16	25	9/9	7/6	7/10
5	12	13	8	1	4	25	144	169	7/10	8/16	16/54
8	15	17	9	2	6	64	225	289	10/24	9/20	19/144
7	24	25	18	1	6	49	576	625	13/36	18/210	13/60

DISCUSSION

In addition to the above, I have (as of 09/29/26) about 100 notebook pages of calculations involving this system of variables, including several unsuccessful attempts to prove the theorem with simple algebra. In those pages, I worked out the variable equations/identities in the "Findings about the system" section, above, wherein I have sometimes referenced the corresponding notebook page numbers.

Throughout the calculations, I have gotten tripped up in algebraic tautologies several times; the logic of how to escape them is not yet always clear to me. Excluding non-integer values of the variables is an even bigger challenge with the way I have gone about this so far. Also there are limitations with my own basic math skills and my understanding of others' work on the theorem.

An AI agent was helpful in gathering information and shaping my understanding of some of the math used in this project, but the core ideas and directions are my own (though they have likely also been pursued previously by others, whose work I have not yet explored).